C++

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```
#include<iostream>
using namespace std;
int main()
{
  char c1,c2;
  int i1,i2;
  float f1,f2;
  c1='H';
  i1=80.56;            /*float converted to int, only 80 assigned to i1  */
  f1=12.6;
  c2=i1;          /*int converted to char */
  i2=f1;          /*float converted to int */
/*Now c2 has the character with ASCII value 80, i2 is assigned value 12
*/
  cout<<"\n c2 = "<<int(c2) << " i2 = "<<i2;    //c2 = 80   i2 = 80
  f2=i1;                /*int converted to float*/
  i2=c1;                /*char converted to int */
  /*Now i2 contains ASCII value of character 'H' which is 72*/
  cout<<"\n f2 = "<<f2<<"  i2 = "<<i2;     //f2 = 80   i2 = 72
```

```
      int x=5,y=2;
      float p,q;
      p=x/y;
      cout<<"\n p = "<<p;     //2
      q=(float)x/y;
      cout<<" q = "<<q;       //2.5


}
```

Output :

??

```
#include<iostream>
#include<climits>
using namespace std;
int main()
{
   unsigned int var = UINT_MAX;
   cout<<"\n var = "<<var;

   var = var + 1;
   cout<<"\n var = "<<var;

   int ivar = INT_MAX;
   cout<<"\n ivar = "<<ivar;

   ivar = ivar + 1;
   cout<<"\n ivar = "<<ivar;

   return 0;
}
```

Output:

```
var = 4294967295
var = 0
ivar = 2147483647
ivar = -2147483648
```

Control Structures in C++:

Condition Structure:

if..else if..else

Loop Structures:

for loop
while loop
do..while loop

Keywords work with loop structure:

break
continue

** Assignments based on condition and loop structures **

Demo Relational Operatros

```
#include<iostream>
using namespace std;
int main( )
{       int a,b ;
        cout<<"Enter values for a and b : ";
        cin>>a ; cin>>b;
        if(a<b)
                cout<< a <<"  is less than  "<<b;
        if(a<=b)
                cout<< a <<"  is less than  or equal to "<<b;
        if(a==b)
                cout<< a <<"  equal to "<<b;
        if(a!=b)
                cout<< a <<"  is not equal to  "<<b;
        if(a>b)
                cout<< a <<"  is greater than  "<<b;
        if(a>=b)
                cout<< a <<"  is greater than or equal to  "<<b;
        return 0;
}
```

The if statement can take any of the following forms:

(a) if (condition) do this ;

(b) if (condition) { do this ; and this ; }

(c) if (condition) do this ; else do this ;

(d) if (condition) { do this ; and this ; } else { do this ; and this ; }

(e) if (condition) do this ; else { if (condition) do this ; else { do this ; and this ; } }

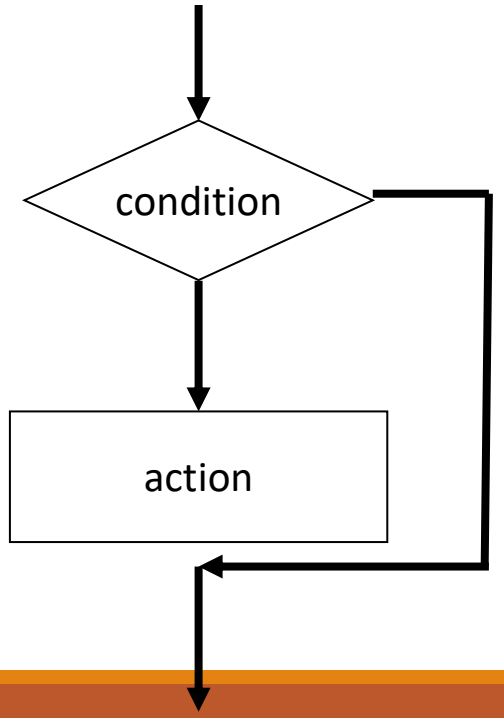
(f) if (condition) { if (condition) do this ; else { do this ; and this ; } } else do this ;

Control Structures: conditional constructs

```
if (x <= 10)
        y = x * x + 5;
```

```
if (x <= 10)
        y = x * x + 5;
        z = (2 * y)/4 ;
```

```
if (x <= 10)      {
        y = x * x + 5;
        z = (2 * y)/4 ;
}
```



1.1 Find biggest number of 3 given numbers :

```
#include<iostream>
using namespace std;
int main()
{
	int a,b,c,big;
	cout<<"Enter three numbers : ";
	cin>>a ; cin>>b ; cin>>c;
	if(a>b)
	{
		if(a>c)
			big=a;
		else
			big=c;
	}
```



```
else
	{
		if(b>c)
			big=b;
		else
			big=c;
	}
	cout<<"\nBiggest number is : "<<big;
	return 0;
}/*End of main()*/
```

Find given year is leap year or not ?

```
#include<iostream>
using namespace std;
int main()
{
	int year;
	cout<<" Enter year : ";
	cin >> year;

	if(year%100 != 0)
	{
		if(year%4 == 0)
			cout<<"\nLeap year ";
		else
			cout<<"\nNot leap ";
	}
```

```
else
	{
		if(year%400 == 0)
			cout<<"\nLeap year ";
		else
			cout<<"\nLeap year ";
	}
	return 0;
}
```

Program to find out the grade of a student when the marks of 4 subjects are given.
The method of assigning grade is -

percentage>=85	grade=A
percentage<85 and percentage>=70	grade=B
percentage<70 and percentage>=55	grade=C
percentage<55 and percentage>=40	grade=D
percentage<40	grade=E

```
if(condition)
  ….
else if(condition)
  …..
else if(condition)
```

```
per=total/4;
        if(per>=85)
                grade='A';
        else if(per>=70)
                grade='B';
        else if(per>=55)
                grade='C';
        else if(per>=40)
                grade='D';
        else
                grade='E';
```

Assignments :

1. Accept a number from the user and check whether the number is positive or negative.
2. Accept a number from the user and check whether the number is odd or even.
3. Accept marks of 5 different subject of a student. Calculate total marks, percentage of that student. Depending upon the percentage display the grade for the student. (Moderate)

   Conditions : percentage more than 80 then A+
   percentage more than 70 then A
   percentage more than 60 then B+
   percentage more than 50 then B
   percentage more than 40 then C
   percentage less than 40 then “Fail”